MAR-ACC-30: Standard Operating Procedure for Resolving Major Optical Transport Failures on Arista Access Nodes at Marathahalli Core

Node: MAR-ACC-30 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_1632

Description

Optical Transport Failures on Arista Access Nodes at Marathahalli Core are primarily caused by SFP/XFP degradation, fiber cuts, or Rx/Tx power loss due to the inherent complexities of Optical Signal Transmission. The Arista Access Node, being a crucial component in the network architecture, relies on the efficient transmission of optical signals between the SFP/XFPs and the fiber optics. When an Optical Transport Failure occurs, it can be attributed to various root causes such as SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. These root causes can be further categorized into physical, environmental, or hardware-related issues. The Arista Access Node's robust architecture and vendor-specific CLI dialect enable the network administrator to diagnose and resolve these failures efficiently. This Standard Operating Procedure outlines the step-by-step approach to resolving Major Optical Transport Failures on Arista Access Nodes at Marathahalli Core.

Pre-requisite Checks

- check: Verify the remote hands have access to the Marathahalli Core site to perform physical checks and troubleshooting on the affected node., status: Completed

- check: Confirm the status of the redundant link to ensure it is functioning properly., status: Completed

- check: Check the environmental conditions at the Marathahalli Core site to rule out any temperature or humidity-related issues., status: Completed

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware sfp details, Expected Output: SFP details, including temperature, voltage, and optical power levels., Description: This command displays detailed information about the SFP/XFP, including temperature, voltage, and optical power levels, which can help diagnose SFP/XFP degradation.

Command: show interface status, Expected Output: Interface status, including Tx/Rx power levels and error counts., Description: This
```

```
command displays the interface status, including Tx/Rx power levels and error counts, which can help diagnose Rx/Tx power loss or fiber cuts.

Command: show log messages, Expected Output: Log messages related to the failure, including alarm codes and timestamps., Description: This command displays log messages related to the failure, including alarm codes and timestamps, which can help identify the root cause of the failure.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP, Command: no hardware sfp 1/1/1, Description: Remove the faulty SFP/XFP and replace it with a new one. Then, reconfigure the SFP/XFP using the 'hardware sfp 1/1/1' command. This step is critical to resolving SFP/XFP degradation failures.

Step 2: Step: Verify the fiber connection, Command: show interface status, Description: Verify that the fiber connection is secure and not damaged. If the fiber connection is damaged, replace it with a new one.

Step 3: Step: Adjust the Rx/Tx power levels, Command: interface 1/1/1, Description: Adjust the Rx/Tx power levels to optimal values. This step is critical to resolving Rx/Tx power loss failures.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface status, including Tx/Rx power levels and error counts., Description: Verify that the interface status indicates normal operation, including optimal Tx/Rx power levels and error counts.

Command: show log messages, Expected Output: No log messages related to the failure., Description: Verify that there are no log messages related to the failure.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP configuration, Command: hardware sfp 1/1/1, Description: Restore the original SFP/XFP configuration to its previous state.

- Step: Verify the fiber connection, Command: show interface status, Description: Verify that the fiber connection is secure and not damaged.

- Step: Adjust the Rx/Tx power levels, Command: interface 1/1/1, Description: Adjust the Rx/Tx power levels to their previous values.